Design and Apply Context-Aware Access Controls Based on User Identity, Device Posture, Location, and Application Risk

Introduction

The project titled **“Design and Apply Context-Aware Access Controls Based on User Identity, Device Posture, Location, and Application Risk”** focuses on improving cybersecurity by implementing intelligent access control mechanisms based on multiple contextual factors. Traditional access control systems usually rely only on usernames and passwords, which are not always sufficient to prevent unauthorized access. In this project, a more advanced approach is considered where access decisions are made based on user identity, the security status of the device being used, the geographic location of the user, and the risk level of the application being accessed. By analyzing these contextual elements, the system can dynamically allow, restrict, or monitor access to sensitive resources. This approach helps organizations strengthen their security posture, reduce the risk of data breaches, and ensure that only trusted users and secure devices can access critical systems. The project also highlights the importance of modern cybersecurity practices such as risk-based authentication, device verification, and location-aware security policies in protecting organizational networks and data.

Stage 1

Title of the project :- Design and Apply Context-Aware Access Controls Based on User Identity, Device Posture, Location, and Application Risk

Overview :-

The main goal of this project is to design and implement a context-aware access control system that improves the security of digital systems. Traditional access control systems only verify user identity through passwords or authentication methods. However, modern cybersecurity threats require more intelligent systems that consider additional contextual information before granting access.

In this project, we propose a context-aware access control model that evaluates multiple factors such as user identity, device posture, geographical location, and application risk level before allowing access to sensitive systems. By analyzing these contextual attributes, the system can make smarter security decisions and prevent unauthorized access attempts.

The system works by collecting contextual information from the user when they attempt to log into an application. This includes the user's identity, the device they are using (such as a trusted company device or personal device), the geographical location from which they are accessing the system, and the sensitivity level of the application being accessed. Based on predefined security policies, the system then determines whether access should be granted, restricted, or denied.

For example, if an authorized user logs in from a trusted device within a secure location, the system will allow full access. However, if the same user tries to log in from an unknown device or an unusual location, the system may restrict access or deny the request completely.

This approach significantly improves cybersecurity by reducing the chances of data breaches, insider threats, and unauthorized system access. Context-aware access control systems are widely used in modern enterprise environments, cloud platforms, and zero-trust security architectures.

The implementation of this project demonstrates how contextual security policies can be applied to protect applications and sensitive resources. The system is implemented using simple programming logic to simulate real-world access control decision-making mechanisms.

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List of Vulnerability Table

S.no	Vulnerability Name	CWE/CVE - No
1	Broken Authentication	CWE-287
2	Improper Access Control	CWE-284
3	Untrusted Device Access	CWE-306
4	Location Spoofing Risk	CWE-346
5	Privilege Escalation	CWE-269
6	Session Hijacking	CWE-384
7	Cross-Site Scripting (XSS)	CWE-79
8	SQL Injection	CWE-89
9	Security Misconfiguration	CWE-16
10	Sensitive Data Exposure	CWE-200

REPORT: -

1. Broken Authentication

Vulnerability Name: Broken Authentication

CWE/CVE: CWE-287

OWASP/SANS Category: OWASP Top 10 – Identification and Authentication Failures

Description:

Broken authentication occurs when authentication mechanisms are incorrectly implemented. Attackers may exploit weak password policies, session management flaws, or exposed authentication tokens to gain unauthorized access.

Business Impact:

Attackers may log in as legitimate users or administrators, gaining access to confidential data and sensitive systems. This may lead to data breaches, financial losses, and reputational damage.

Vulnerability Path:

http://example.com/login

Vulnerability Parameter:

http://example.com/login?username=admin&password=admin

Steps to Reproduce:

1. Open the login page.
2. Enter weak or default credentials.
3. Submit the login form.
4. The system grants access without proper validation.

2. Improper Access Control

Vulnerability Name: Improper Access Control

CWE/CVE: CWE-284

OWASP/SANS Category: OWASP Top 10 – Broken Access Control

Description:

Improper access control occurs when users can access resources or perform actions that they are not authorized to perform due to weak permission checks.

Business Impact:

Unauthorized users may access confidential information or administrative functions, potentially causing system compromise or data leakage.

Vulnerability Path:

http://example.com/admin

Vulnerability Parameter:

http://example.com/admin?role=user

Steps to Reproduce:

1. Log in as a normal user.
2. Modify the URL to access an admin page.
3. If the system does not check permissions, the page loads.

3. Untrusted Device Access

Vulnerability Name: Untrusted Device Access

CWE/CVE: CWE-306

OWASP/SANS Category: OWASP – Security Misconfiguration

Description:

This vulnerability occurs when systems allow login from unverified or unknown devices without validating device trust or security posture.

Business Impact:

Attackers using compromised or unauthorized devices may gain access to sensitive applications, increasing the risk of malware infection and data theft.

Vulnerability Path:

http://example.com/device-login

Vulnerability Parameter:

http://example.com/device-login?device=unknown

Steps to Reproduce:

1. Attempt login from an unregistered device.
2. Submit login credentials.
3. If the system does not check device trust, access is granted.

4. Location Spoofing Risk

Vulnerability Name: Location Spoofing Risk

CWE/CVE: CWE-346

OWASP/SANS Category: OWASP – Security Misconfiguration

Description:

Location spoofing occurs when attackers manipulate their network location information to bypass location-based access restrictions.

Business Impact:

Attackers may access restricted services from unauthorized regions, increasing the likelihood of cyberattacks and data compromise.

Vulnerability Path:

http://example.com/location-access

Vulnerability Parameter:

http://example.com/location-access?location=USA

Steps to Reproduce:

1. Use a VPN or proxy server.
2. Change the IP address location.
3. Access the restricted application.

5. Privilege Escalation

Vulnerability Name: Privilege Escalation

CWE/CVE: CWE-269

OWASP/SANS Category: OWASP – Broken Access Control

Description:

Privilege escalation happens when a user gains higher permissions than intended, allowing them to perform administrative tasks or access restricted data.

Business Impact:

Attackers may gain full control of the system, modify data, disable security protections, or compromise the entire infrastructure.

Vulnerability Path:

http://example.com/user/profile

Vulnerability Parameter:

http://example.com/user/profile?role=admin

Steps to Reproduce:

1. Log in as a normal user.
2. Modify role parameter in the request.
3. The system assigns admin privileges.

6. Session Hijacking

Vulnerability Name: Session Hijacking

CWE/CVE: CWE-384

OWASP/SANS Category: OWASP – Identification and Authentication Failures

Description:

Session hijacking occurs when attackers steal or manipulate session IDs to impersonate legitimate users and access their accounts.

Business Impact:

Attackers may access private data, perform unauthorized transactions, or compromise user accounts.

Vulnerability Path:

http://example.com/session

Vulnerability Parameter:

http://example.com/session?id=123456

Steps to Reproduce:

1. Capture session cookie using a network tool.
2. Replace the cookie in the browser.
3. Access the account using the stolen session.

7. Cross-Site Scripting (XSS)

Vulnerability Name: Cross-Site Scripting (XSS)

CWE/CVE: CWE-79

OWASP/SANS Category: OWASP Top 10 – Injection

Description:

Cross-Site Scripting is a vulnerability where attackers inject malicious scripts into web pages viewed by other users.

Business Impact:

Attackers may steal cookies, session tokens, or personal information, leading to account compromise and data leakage.

Vulnerability Path:

http://example.com/search

Vulnerability Parameter:

http://example.com/search?q=<script>alert('hacked')</script>

Steps to Reproduce:

1. Open the search page.
2. Enter a script payload in the search field.
3. Submit the form.
4. The browser executes the injected script.

8. SQL Injection

Vulnerability Name: SQL Injection

CWE/CVE: CWE-89

OWASP/SANS Category: OWASP Top 10 – Injection

Description:

SQL injection occurs when attackers insert malicious SQL queries into input fields, allowing them to manipulate database queries.

Business Impact:

Attackers may access, modify, or delete sensitive database information, causing serious security and operational issues.

Vulnerability Path:

http://example.com/login

Vulnerability Parameter:

http://example.com/login?username=admin'--&password=test

Steps to Reproduce:

1. Open login page.
2. Enter SQL payload in the username field.
3. Submit login request.
4. The system bypasses authentication.

9. Security Misconfiguration

Vulnerability Name: Security Misconfiguration

CWE/CVE: CWE-16

OWASP/SANS Category: OWASP Top 10 – Security Misconfiguration

Description:

Security misconfiguration occurs when systems are deployed with default settings, unnecessary services, or incorrect security settings.

Business Impact:

Improper configuration can expose systems to attackers, allowing them to exploit vulnerabilities and gain unauthorized access.

Vulnerability Path:

http://example.com/admin-panel

Vulnerability Parameter:

http://example.com/admin-panel?debug=true

Steps to Reproduce:

1. Access admin panel using default credentials.
2. Enable debug mode.
3. Sensitive information becomes visible.

10. Sensitive Data Exposure

Vulnerability Name: Sensitive Data Exposure

CWE/CVE: CWE-200

OWASP/SANS Category: OWASP Top 10 – Sensitive Data Exposure

Description:

Sensitive data exposure occurs when confidential information such as passwords, credit card details, or personal data is not properly protected.

Business Impact:

Data breaches can lead to identity theft, financial fraud, regulatory penalties, and damage to organizational reputation.

Vulnerability Path:

http://example.com/user-data

Vulnerability Parameter:

http://example.com/user-data?id=1001

Steps to Reproduce:

1. Access the data endpoint.
2. Intercept network traffic.
3. Observe sensitive data transmitted without encryption.

Stage 2

Overview :-

Nessus is one of the most widely used vulnerability assessment tools in the field of cybersecurity. It is developed by the cybersecurity company Tenable and is designed to identify security vulnerabilities, configuration issues, and potential threats in computer systems, networks, and web applications.

The main purpose of Nessus is to help organizations detect security weaknesses before attackers exploit them. It works by scanning target systems and comparing them against a large database of known vulnerabilities and security misconfigurations. Nessus uses thousands of plugins that test systems for known vulnerabilities, outdated software versions, and configuration issues.

During a vulnerability scan, Nessus analyzes open ports, running services, and application behaviors. Based on its findings, it generates a detailed report that categorizes vulnerabilities based on severity levels such as Critical, High, Medium, Low, and Informational. This helps security teams prioritize which vulnerabilities need immediate attention.

Nessus supports multiple types of scans including network scans, web application scans, compliance checks, and malware detection. It can identify issues such as weak passwords, missing patches, insecure services, and exposure of sensitive data.

One of the major advantages of Nessus is its large plugin database which is regularly updated to detect newly discovered vulnerabilities. Security professionals often use Nessus as part of their vulnerability management process to continuously monitor and improve system security.

In this project, Nessus is used to simulate vulnerability scanning on a target system. The scan results help identify security weaknesses that could be exploited by attackers. These vulnerabilities are then analyzed, documented, and recommendations are provided to mitigate the risks.

Target website — http://testfire.net

Target ip address:-

65.61.137.117

List of vulnerability

S.No	Vulnerability Name	Severity	Plugin
1	Cross-Site Scripting (XSS)	High	10815
2	SQL Injection	Critical	11219
3	Security Misconfiguration	Medium	51192
4	Sensitive Data Exposure	High	57582
5	Broken Authentication	Medium	35291
6	Improper Access Control	High	42056
7	Session Hijacking	High	26194
8	Privilege Escalation	Critical	33850
9	Untrusted Device Access	Medium	45590
10	Location Spoofing Risk	Low	58745

REPORT:-

1. Vulnerability Name: Cross-Site Scripting (XSS)

Severity: High

Plugin: 10815

Port: 80

Description:

Cross-Site Scripting (XSS) is a web security vulnerability that allows attackers to inject malicious scripts into web pages viewed by other users. These scripts run in the victim's browser and can steal cookies, session tokens, or other sensitive information.

Solution:

• Validate and sanitize all user inputs before processing them.

• Use output encoding to prevent malicious scripts from executing.

• Implement Content Security Policy (CSP) to restrict script execution.
• Regularly test the application using security scanning tools.
Business Impact:
If Cross-Site Scripting vulnerabilities are exploited, attackers can inject malicious scripts that execute in the browsers of legitimate users. This can allow attackers to steal session cookies, login credentials, or other sensitive user information. In some cases, attackers may impersonate users and perform unauthorized actions within the application. Such attacks may lead to loss of user trust, exposure of confidential data, and reputational damage for the organization. If sensitive systems are affected, the organization may also face regulatory penalties, financial losses, and long-term damage to its brand and customer relationships.

2. Vulnerability Name: SQL Injection
Severity: Critical
Plugin: 11219
Port: 80
Description:
SQL Injection occurs when attackers insert malicious SQL queries into input fields, allowing them to manipulate the database and retrieve or modify sensitive data.
Solution:
• Use parameterized queries or prepared statements for database operations.
• Validate and sanitize all user inputs before processing them.
• Implement least privilege access control for database users.
• Regularly update and patch database management systems.
Business Impact:
SQL Injection attacks can have severe consequences for an organization because they allow attackers to interact directly with the database. If successfully exploited, attackers may retrieve sensitive information such as usernames, passwords, financial records, or personal customer data. They may also modify or delete important data, affecting business operations. Such incidents can lead to data breaches, legal liabilities, and loss of customer trust. Organizations may face regulatory penalties and financial losses due to compromised data security and disruption of services.

3. Vulnerability Name: Security Misconfiguration

Severity: Medium

Plugin: 51192

Port: 443

Description:

Security misconfiguration occurs when systems are deployed with insecure default settings or improper configurations that expose vulnerabilities.

Solution:

• Remove unnecessary services and features from the system.

• Change default usernames and passwords immediately.

• Regularly update and patch operating systems and applications.

• Perform regular security audits and configuration reviews.

Business Impact:

Security misconfigurations can expose systems to attackers by leaving default settings, open ports, or unnecessary services active. These weaknesses can be exploited to gain unauthorized access to systems or sensitive information. Attackers may use misconfigured servers to launch further attacks or gain deeper access into the network infrastructure. This may result in data breaches, service disruptions, and operational downtime. Additionally, organizations may suffer reputational damage and financial losses due to compromised systems and loss of customer confidence.

4. Vulnerability Name: Sensitive Data Exposure

Severity: High

Plugin: 57582

Port: 443

Description:

Sensitive Data Exposure occurs when confidential information such as passwords, financial details, or personal data is not properly protected.

Solution:

• Use strong encryption techniques to protect sensitive data.

• Implement secure communication protocols such as HTTPS.

• Store passwords using secure hashing algorithms.
• Limit access to sensitive data using strict access controls.
Business Impact:
Sensitive data exposure can have serious consequences for organizations because confidential information may be accessed by unauthorized users. If attackers obtain personal data, financial records, or login credentials, they can misuse the information for fraud, identity theft, or other malicious activities. This can lead to legal consequences, financial penalties, and loss of customer trust. Organizations may also experience reputational damage and reduced customer confidence if sensitive information is leaked or misused.

5. Vulnerability Name: Broken Authentication
Severity: Medium
Plugin: 35291
Port: 80
Description:
Broken authentication occurs when authentication mechanisms are poorly implemented, allowing attackers to bypass login systems or guess user credentials.
Solution:
• Implement strong password policies for users.
• Enable multi-factor authentication (MFA).
• Use secure session management techniques.
• Monitor login attempts and detect suspicious activities.
Business Impact:
Broken authentication vulnerabilities allow attackers to bypass authentication mechanisms and gain unauthorized access to user accounts or administrative systems. Once attackers gain access, they may view or modify sensitive information, impersonate legitimate users, and perform malicious actions. This can result in data breaches, service disruptions, and financial losses. Organizations may face regulatory issues and reputational damage if customer data is compromised due to weak authentication systems.

6. Vulnerability Name: Improper Access Control

Severity: High

Plugin: 42056

Port: 80

Description:

Improper access control occurs when an application does not properly enforce restrictions on what authenticated users are allowed to do. Attackers may exploit this vulnerability to access unauthorized data or perform restricted operations.

Solution:

• Implement role-based access control (RBAC) to manage permissions.

• Validate user authorization before granting access to resources.

• Restrict access to sensitive data based on user roles.

• Perform regular security audits to detect permission issues.

Business Impact:

Improper access control can allow unauthorized users to view or modify sensitive data within the system. Attackers may exploit this weakness to access confidential information such as user records, financial data, or administrative functions. In severe cases, attackers may modify system configurations or delete important data, causing operational disruptions. Such incidents can lead to loss of customer trust, financial damage, and potential legal consequences. Organizations may also experience reputational harm and regulatory penalties if sensitive information is exposed due to weak access control mechanisms.

7. Vulnerability Name: Session Hijacking

Severity: High

Plugin: 26194

Port: 80

Description:

Session hijacking occurs when attackers steal or manipulate a valid session ID to gain unauthorized access to a user’s account. This allows attackers to impersonate legitimate users.

Solution:

• Use secure cookies and enable HTTPS for all communications.
• Implement session timeout mechanisms to limit session duration.
• Regenerate session IDs after successful authentication.
• Monitor and detect suspicious session activities.

Business Impact:

Session hijacking can allow attackers to gain unauthorized access to user accounts by stealing valid session identifiers. Once attackers obtain a session ID, they can impersonate the legitimate user and perform actions within the application. This may include accessing confidential information, modifying account settings, or performing unauthorized transactions. Such incidents can lead to financial losses, compromise of sensitive data, and damage to the organization's reputation. Additionally, users may lose trust in the application's security, resulting in reduced customer confidence and potential legal liabilities.

8. Vulnerability Name: Privilege Escalation

Severity: Critical

Plugin: 33850

Port: 443

Description:

Privilege escalation occurs when attackers exploit system vulnerabilities to gain higher access privileges than intended. This allows them to perform administrative actions within the system.

Solution:

• Implement strict role-based access control policies.
• Regularly update and patch system software.
• Monitor system activities for unusual privilege changes.
• Limit administrative privileges to authorized users only.

Business Impact:

Privilege escalation vulnerabilities can have severe consequences for organizations because attackers can gain administrative or root-level access to systems. With elevated privileges, attackers may control system operations, access sensitive data, modify system configurations, or disable security controls. This could result in complete system compromise and disruption of business operations. Organizations may suffer financial losses, data breaches, and reputational

damage. If critical systems are affected, the organization may also face regulatory penalties and legal consequences due to failure to protect sensitive information.

9. Vulnerability Name: Untrusted Device Access
Severity: Medium
Plugin: 45590
Port: 80
Description:
Untrusted device access occurs when applications allow login from devices that are not verified or trusted by the system. Attackers may use compromised devices to gain access.
Solution:
• Implement device authentication and verification mechanisms.
• Allow access only from trusted or registered devices.
• Monitor login attempts from unknown devices.
• Use multi-factor authentication for additional security.
Business Impact:
Allowing access from untrusted devices increases the risk of unauthorized system access. Attackers using compromised or insecure devices may gain entry to the application and attempt to steal sensitive data or perform malicious activities. This may lead to exposure of confidential information and disruption of normal operations. Organizations may face data breaches, financial losses, and reputational damage if attackers successfully exploit this vulnerability. Implementing strong device verification mechanisms is essential to maintain system security and prevent unauthorized access.

10. Vulnerability Name: Location Spoofing Risk
Severity: Low
Plugin: 58745
Port: 80

Description:

Location spoofing occurs when attackers manipulate their network location or IP address to bypass geographic restrictions placed on a system.

Solution:

- Implement IP verification and geolocation monitoring.
- Detect and block suspicious VPN or proxy usage.
- Use additional authentication for unusual login locations.
- Monitor login patterns and generate alerts for anomalies.

Business Impact:

Location spoofing attacks allow unauthorized users to bypass geographic access restrictions and gain entry to restricted systems. Attackers may disguise their real location using VPNs or proxy servers to appear as if they are accessing the system from a trusted region. This can lead to unauthorized access to sensitive applications and data. If attackers successfully exploit this vulnerability, organizations may experience data breaches, system compromise, and loss of sensitive information. Continuous monitoring and location verification are necessary to reduce the risk of such attacks.

Stage 3

Report

Title :- Security Monitoring and Threat Detection using SOC and SIEM

SOC (Security Operations Center)

A Security Operations Center (SOC) is a centralized unit responsible for monitoring, detecting, analyzing, and responding to cybersecurity incidents in an organization. The SOC team continuously monitors network traffic, system logs, and security alerts to identify potential threats and vulnerabilities. Security analysts working in a SOC use various tools to detect malicious activities such as unauthorized access, malware attacks, and data breaches. The main goal of a SOC is to maintain the security of the organization's systems and data by responding quickly to security incidents and minimizing damage.

SOC Cycle

The SOC cycle represents the continuous process followed by security teams to manage cybersecurity threats. The cycle generally includes monitoring, detection, investigation, response, and improvement. First, the SOC team monitors network activity and system logs to detect suspicious behavior. Once a potential threat is identified, analysts investigate the issue to understand its nature and severity. After confirmation, appropriate actions are taken to respond to the threat and mitigate its impact. Finally, the team reviews the incident and improves security measures to prevent similar attacks in the future.

SIEM

Security Information and Event Management (SIEM) is a security solution that collects and analyzes log data from various systems and devices within a network. SIEM tools help security teams detect threats by correlating events from multiple sources such as servers, firewalls, and applications. By analyzing this data, SIEM systems can identify abnormal patterns and generate alerts for potential security incidents. SIEM plays an important role in modern cybersecurity operations because it provides centralized monitoring, improves threat detection, and helps organizations respond quickly to security incidents.

SIEM Cycle

The SIEM cycle involves collecting, analyzing, and responding to security events. First, log data is collected from different sources such as servers, applications, and network devices. The collected data is then processed and analyzed to identify suspicious activities. The SIEM system correlates multiple events to detect potential threats and generates alerts for security teams. After receiving alerts, analysts investigate the incidents and take appropriate actions to mitigate risks. This cycle continues continuously to maintain the security of the organization's infrastructure.

MISP

MISP (Malware Information Sharing Platform) is an open-source threat intelligence platform used to share information about cyber threats. It allows organizations to collaborate and exchange data about malware, vulnerabilities, and attack patterns. By sharing threat intelligence, organizations can better understand emerging threats and improve their security defenses. MISP helps security teams identify indicators of compromise (IOCs) and respond quickly to cyber attacks. It plays a crucial role in improving cybersecurity collaboration between organizations.

Your College Network Information

The college network infrastructure typically includes multiple computers, servers, routers, and internet-connected devices that support academic and administrative operations. Students and staff use the network to access learning management systems, online resources, and communication platforms. The network may include Wi-Fi access points, local servers, and cloud services. Proper network security measures such as firewalls, antivirus systems, and access control mechanisms are necessary to protect the network from cyber threats and unauthorized access.

How You Think You Deploy SOC in Your College

To deploy a Security Operations Center in a college environment, security monitoring systems should be installed to track network activities and detect suspicious behavior. Security tools such as SIEM platforms can collect logs from servers, routers, and firewalls within the campus network. A dedicated security team can monitor these logs and investigate potential threats. Implementing access control policies, vulnerability scanning, and regular security audits can further strengthen the college's cybersecurity infrastructure.

Threat Intelligence

Threat intelligence refers to the collection and analysis of information related to potential cyber threats. It helps organizations understand attacker behavior, emerging vulnerabilities, and potential attack techniques. By using threat intelligence, security teams can proactively identify risks and improve their defense strategies. Threat intelligence sources include security reports, vulnerability databases, and threat sharing platforms. This information helps organizations prepare for attacks and respond effectively to cybersecurity incidents.

Incident Response

Incident response is the process of identifying, managing, and resolving cybersecurity incidents such as data breaches, malware infections, or unauthorized access attempts. The incident response process typically includes detection, analysis, containment, eradication, and recovery. Security teams investigate incidents to determine their cause and impact, then take steps to eliminate the threat and restore normal operations. Effective incident response helps minimize damage and ensures that similar incidents are prevented in the future.

QRadar & Understanding About Tool

IBM QRadar is a powerful SIEM tool used for security monitoring and threat detection. It collects log data from various systems and analyzes it to identify potential security threats. QRadar uses advanced analytics and correlation techniques to detect suspicious activities in real time. Security analysts can use QRadar dashboards to monitor network traffic, investigate incidents, and respond quickly to cyber threats. By providing centralized visibility and automated threat detection, QRadar helps organizations strengthen their cybersecurity operations.

Conclusion

Stage 1

From Stage 1, we understood the importance of web application security testing and the identification of common vulnerabilities such as SQL injection, cross-site scripting, and broken authentication. Understanding these vulnerabilities helps developers design more secure applications and prevent cyber attacks.

Stage 2

In Stage 2, we learned how vulnerability scanning tools such as Nessus can identify security weaknesses in systems and applications. The scan results provide detailed information about vulnerabilities, their severity levels, and recommended solutions.

Stage 3

In Stage 3, we understood the role of SOC and SIEM systems in monitoring security events and detecting threats. Tools like IBM QRadar help security teams analyze log data and respond quickly to cybersecurity incidents.

Future Scope

Stage 1

Future web application testing will involve advanced automated tools and AI-based security analysis to detect vulnerabilities more efficiently.

Stage 2

Future vulnerability scanning processes will integrate artificial intelligence and machine learning to identify security risks more accurately.

Stage 3

SOC and SIEM systems will continue to evolve with advanced threat detection technologies and automated incident response capabilities.

Topics Explored

- Web Application Security
- Vulnerability Assessment
- Security Operations Center (SOC)
- Security Information and Event Management (SIEM)
- Threat Intelligence
- Incident Response

Tools Explored

- Nessus
- IBM QRadar
- Malware Information Sharing Platform

THE END